

Market Study & Feasibility Report

Predicting railway level-crossing gate closures along a driving route, from live train positions. An honest assessment — usefulness, defensibility, and real-world adoption.

The question this report answers: it's an obvious, valuable idea — so **why hasn't anyone already built it**, and **will people actually use it?**

Four questions, answered straight: (1) How can RGT actually be useful? (2) Why hasn't everyone already built it? (3) Is it *really* useful? (4) Will people *actually* use it?

~18,000

MANNED CROSSINGS
STILL LIVE

63%

OF RAILWAY DEATHS
AT CROSSINGS

~120

VEHICLES QUEUED
PER 10-MIN CLOSURE

>3 min

WAIT THAT TRIGGERS
RISKY BEHAVIOUR

1 How RGT can be useful

The job-to-be-done: **“Don't make me get stuck at a closed railway gate.”** RGT predicts, *before* you commit to a route, which level crossings on it will be shut and for how long — so you can re-route, leave later, or simply not be surprised.

Who it helps, ordered by how much the pain is worth money:

- **Fleets & time-critical drivers** — logistics, delivery, school buses, **ambulances**, cab aggregators. A blocked crossing is measurable lost time and fuel, and a dispatcher routing many vehicles compounds the value. This is where the pain has a price tag.
- **Daily commuters** on corridors with busy crossings — the classic “I left on time and still lost 12 minutes at the gate.”
- **Traffic authorities / city planners** — aggregate closure and congestion data around crossings informs signal timing and ROB/RUB prioritisation.

Why it's possible at all: India runs **~18,000 manned level crossings**, ~54% interlocked with signals (closures are deterministic from train movement), and live train positions are now obtainable via the **RailRadar API**. The raw materials exist; nobody has assembled them into a driver-facing predictor.

2 Why hasn't everyone already built it?

This is the decisive question — “obvious + valuable + unbuilt” usually hides a barrier. Here are the real ones, which are also RGT's moat.

(a) The live train data was genuinely hard to get — until recently. There is no official public API for live Indian train data. NTES partner feeds are limited to approved partners; IRCTC sits behind captcha/OTP/anti-bot controls and a ToS that forbids scraping. Anyone attempting this resorted to brittle scraping (exactly what RGT v1 did with Selenium — and why it kept breaking). Paid aggregators like RailRadar are what make a *reliable* version newly feasible. The barrier didn't vanish; it became a ₹1,000/mo line item. Most earlier attempts died here.

(b) The level-crossing dataset doesn't exist as clean open data. You need precise gate coordinates, which gates are interlocked, and how each maps to bracketing stations — manual, hyperlocal, unglamorous data work, corridor by corridor. It doesn't scale from a laptop and it's boring, so it rarely gets done. Whoever builds it owns the moat.

(c) It's a cross-domain orphan. The problem sits *between* railways and roads. Train apps (RailYatri ~75M users, Where Is My Train, ixigo, ConfirmTkt) serve passengers; map apps own road routing but have no train-occupancy data. The valuable intersection — train data → road gate closure → your car's ETA — is nobody's core business.

(d) Incumbent incentives point elsewhere. Train apps monetise passenger intent (tickets/ads); a driver avoiding a crossing isn't their funnel. For Google Maps it's a low-priority, single-country, hyperlocal feature. Rational for them to skip — leaving room for a focused player.

(e) The government's official answer is concrete, not software. India is physically eliminating crossings via Rail-Over/Under-Bridges and interlocking — a ~₹41,200 crore, multi-decade programme. But it's slow and leaves **~18,000 crossings live for the next 10–20 years** — a long interim window for a software mitigation. (The policy literature has explicitly proposed this exact app — strong validation, but also a hint the state could build it.)

(f) Prediction is harder than a demo makes it look. Nearest-station ≠ the exact minute a specific gate shuts. Trains skip halts, gates close 5–10 min early, interlocking and double-tracking vary. Turning “a train is near” into “this gate, this window, this confidence” — and *proving* it — is real engineering most prototypes never validate.

NET

It's unbuilt not because it's a bad idea, but because it requires stitching hard-to-get data with a dataset nobody wants to build, across two domains neither incumbent owns, for a problem the state is “solving” on a 20-year timeline. That combination is a moat for a focused builder.

3 Is it *really* useful?

The case for

- Large, quantified pain: ~120 vehicles queued per 10-min closure; **41% of accidents and 63% of deaths** on the railways stem from crossing desperation; waits >3 min trigger risky behaviour.
- It's **preventive** — re-routing before you're stuck beats the status quo, where Google Maps routes you straight into the gate.

The honest case against

- **Low value once you're there** — if you can see the gate is down, the app told you nothing. The value is upstream, at route-choice time.
- **Many crossings, short waits** — lots of closures are 2–4 min, below the re-route threshold. The high-value cases are a subset.
- **Slowly shrinking market** — every ROB built removes a crossing. A “win the next decade” product, not a forever one.

VERDICT

Genuinely useful for a **focused segment** (busy corridors + fleets / time-critical drivers); oversold as a mass-market everyday consumer app. Aim where being stuck actually costs money.

4 Will people *actually* use it?

Separate *useful* from *used* — this is where most good ideas die.

- **Standalone consumer app = low frequency.** You'd only open it near a crossing — weak daily habit, hard retention, costly to acquire users for a few-times-a-week utility. As pure B2C, adoption is an uphill climb.
- **It gets used when it's ambient, not opened:**
 - *Embedded in navigation* — a gate-closure layer inside routing you already do. The real consumer path: become a light nav app for its corridors, or license the layer.
 - *Push, not pull* — “Leave 5 min later; Vikhroli gate shuts 7:45–7:55” for saved commutes turns a low-frequency tool into a passive daily value-add.
 - *Fleet dashboards* — drivers adopt nothing; the dispatcher decides once and routes everyone. Highest realistic usage, least adoption friction.
- **Trust gate** — people abandon a predictor that's wrong twice. Usage is gated on a published, measured accuracy number.

HONEST ANSWER

Yes — but not as a standalone consumer download. It will be used as a fleet/B2B tool (one buyer onboards many users) and a notification-first commute layer, ideally inside an existing navigation flow. A cold standalone Play Store app is the *least* likely path to real usage.

Bottom line

Useful?	Yes — for a focused, high-value segment, with quantified pain and a clear gap.
Why unbuilt?	Hard-to-get data + an unbuilt dataset + a cross-domain orphan problem + a state solving it slowly in concrete. Those barriers are now partly lifted (RailRadar); the rest (the gate dataset + proven accuracy) are exactly what a focused team can own.
Will it be used?	As a fleet/B2B tool and a notification-first commute layer — plausibly yes. As a standalone consumer app — probably not. Build for the segment that pays for time.

The deciding factor

Proven prediction accuracy on one real corridor — “X% accurate within 5 minutes.” Usefulness, defensibility, adoption and monetisation all rest on that single number. It is the whole ballgame.

RECOMMENDED MODEL

Open-core / freemium consumer for adoption and accuracy data (the real moat), with revenue led by **B2B fleets / logistics**, then B2G traffic analytics, then premium consumer. Avoid leading with consumer ads alone.

Sources.

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